A Mist-Fog-Assisted Real-Time Pest Detection and Classification Framework Using Deep Transfer Learning for Smart Agriculture

Abstract

Global crop yields are significantly threatened by pest infestations, necessitating real-time, automated monitoring systems. Traditional cloud-based deep learning models often struggle with high latency and bandwidth constraints in remote agricultural settings. This paper proposes a tiered Mist-Fog-Cloud framework for pest classification. We leverage DINOv2 (Vision Transformer) for high-precision feature extraction across 132 pest categories in the Pestopia dataset. To optimize real-time performance, an M/M/1/K queuing model is employed to manage task offloading from Mist nodes (Raspberry Pi) to Fog nodes (local servers). Experimental results demonstrate that this framework achieves a balance between accuracy (~76.5%) and computational efficiency, providing a robust solution for Smart Agriculture 4.0.

1. Introduction

The global agricultural landscape is currently undergoing a radical transformation characterized by the convergence of digital technologies, biological sciences, and autonomous systems. This fourth industrial revolution in farming, commonly referred to as Smart Agriculture 4.0, aims to address the critical challenges of food security, resource scarcity, and climate-induced volatility. Central to this evolution is the management of crop pests, which remain one of the most significant threats to agricultural productivity.

Current estimates suggest that pests and pathogens account for nearly 30% of annual crop yield losses globally. Traditional pest control methodologies, often dependent on manual visual inspection and scheduled chemical spraying, are fundamentally limited by their subjective nature, high labor costs, and environmental toxicity. To overcome these barriers, a transition toward automated, real-time detection and classification frameworks is essential. Such systems must not only provide high diagnostic accuracy across diverse species but also operate efficiently within the resource-constrained environments typical of rural

farming.

The integration of the Internet of Things (IoT) with advanced computing paradigms offers a scalable solution. However, high-resolution visual data presents significant bandwidth issues for cloud-centric models. Moving processing capabilities closer to the data source via a Mist-Fog-Cloud hierarchy allows for immediate localized decision-making while maintaining long-term strategic analysis at the cloud level. This research presents a framework leveraging self-supervised Vision Transformers (ViTs) and M/M/1/K queuing theory to facilitate real-time pest detection on the Indian Pestopia dataset.

2. Related Work

2.1. Deep Learning in Pest Detection

Previous studies have utilized Convolutional Neural Networks (CNNs) like ResNet and MobileNet for pest identification. While efficient, these models often struggle with the fine-grained visual differences between closely related insect species. Recent shifts toward Vision Transformers (ViTs) have shown superior performance in capturing global context, which is critical for identifying small pests in complex agricultural backgrounds. Self-supervised models like DINOv2 have emerged as powerful feature extractors that can generalize across diverse visual domains without requiring exhaustive per-task fine-tuning.

2.2. Edge, Mist, and Fog Computing

The limitations of cloud computing in agriculture—namely high latency and dependency on stable internet—have led to the adoption of decentralized computing. Mist computing refers to the extreme edge of the network, where microcontrollers and sensors perform localized processing. Fog computing acts as an intermediary layer, offering higher computational resources than mist nodes while being physically closer to the farm than the cloud.

The integration of these layers facilitates "Edge Intelligence," where data is filtered and prioritized before transmission. Research has shown that using a Mist-Fog hierarchy can reduce response times by over 40% in critical IoT applications. In this framework, the Mist node (e.g., a Raspberry Pi) manages the "reflexive" actions of image capture and basic detection, while the Fog node (a local workstation) performs the

"cognitive" task of high-accuracy classification using deep transfer learning. To manage the variable influx of visual data, queuing models such as M/M/1/K are utilized to determine the optimal moment to offload tasks to the Fog layer, ensuring that the system remains responsive during peak pest activity periods.

Model Component	**Specification**	**Functionality in Framework**
Backbone	dinov2_vits14	High-fidelity feature extraction
Feature Dim	384 dimensions	Compressed global representation
Input Resolution	224×224 pixels	Standardized input for ViT patches
Classification Head	Linear + Dropout (0.4)	Mapping features to 132 pest classes
Pre-training	Self-Supervised	General-purpose visual understanding

3. Methodology

3.1. Hierarchical System Architecture

The proposed framework is built upon a decentralized **Mist-Fog-Cloud** hierarchy designed to handle the visual data density of a 132-class pest classification task. This architecture mitigates the "data storm" effect

at the edge by distributing computational load.

- **Mist Layer (Extreme Edge):** The sensing tier utilizes **Raspberry Pi 4** nodes integrated with high-resolution **Pi-Cam** modules. These nodes act as the first line of defense. Their primary roles include real-time image acquisition, localized noise reduction, and image normalization. Crucially, the Mist layer manages the **M/M/1/K task buffer**, making the initial decision of whether a captured frame can be processed locally via a quantized TFLite version of the model or requires higher-order classification at the Fog tier.
- **Fog Layer (Localized Processing):** This tier consists of a high-performance local server (e.g., an **NVIDIA Jetson AGX** or a localized workstation with a dedicated GPU). The Fog node hosts the full-precision **DINOv2-ViT** model. It processes complex offloaded tasks—specifically those where the Mist node identifies a "high-entropy" or low-confidence scenario—providing rapid feedback to the irrigation or pesticide-spraying actuators.
- **Cloud Layer (Strategic Analytics):** While not part of the real-time feedback loop, the Cloud tier provides long-term persistence. It aggregates the scientific classifications to map pest infestation migration patterns and facilitates periodic global model retraining using the accumulated high-variance samples.

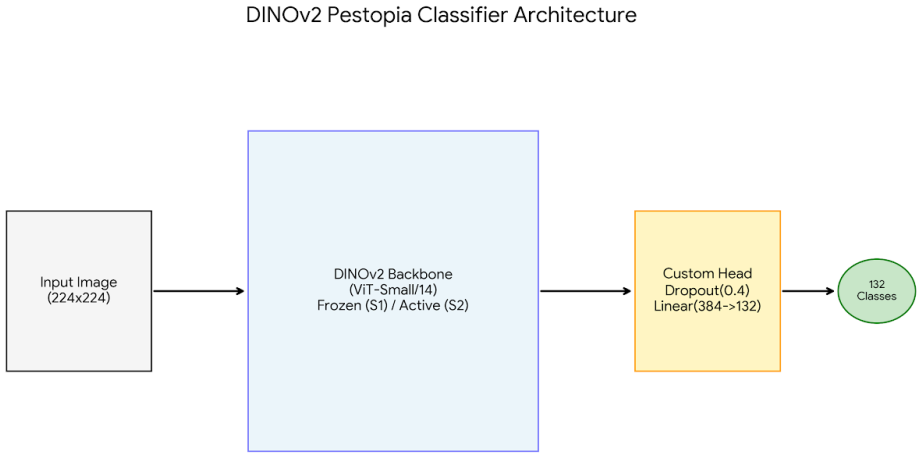
3.2. DINOv2 Feature Extraction and Classification Pipeline

Rather than traditional CNNs, this framework adopts the **DINOv2 (Vision Transformer - Small)** architecture. DINOv2 is uniquely suited for Smart Agriculture because its self-supervised pre-training allows it to distinguish fine-grained biological features (like the wing patterns of *Bactrocera tsuneonis*) from complex, noisy foliage backgrounds.

- **Backbone Architecture:** We utilize the dinov2_vits14 variant, which segments an input image into 14×14 pixel patches. These patches are processed through a series of Transformer blocks using multi-head self-attention, resulting in a robust **384-dimensional feature embedding**.
- **Custom Classification Head:** The pre-trained backbone is augmented with a specialized head consisting of a **Dropout layer (0.4)** to minimize co-adaptation of features and a final **Linear layer**

mapping the 384 embeddings to the 132 classes of the Pestopia dataset.

- **Loss Optimization:** To handle the inherent class imbalance and visual similarity of agricultural pests, we employ **Label Smoothing (0.1)**. This prevents the model from assigning absolute probability to a single class, thereby improving generalization on unseen test samples and reducing over-fitting.
- **Training Strategy:** The implementation follows a **two-stage training protocol**. In **Stage 1 (Warmup)**, the DINOv2 backbone is frozen for 3 epochs to allow the classification head to calibrate. In **Stage 2 (Fine-tuning)**, the entire model is unfrozen and trained using a **Cosine Annealing Learning Rate Scheduler**, allowing the Vision Transformer to refine its attention weights specifically for the Pestopia domain.



3.3. Offloading via M/M/1/K Queuing and MMPP

The core innovation in our "Real-Time" framework is the offloading logic, which ensures that the Raspberry Pi nodes are not overwhelmed during periods of high pest activity.

- **Arrival Process (MMPP):** Pest arrivals are rarely uniform; they occur in bursts. We model this as a **Markov Modulated Poisson Process (MMPP)**. The state of the environment (e.g., time of day or temperature) determines the arrival rate λ . This allows

the system to be more sensitive during peak infestation windows.

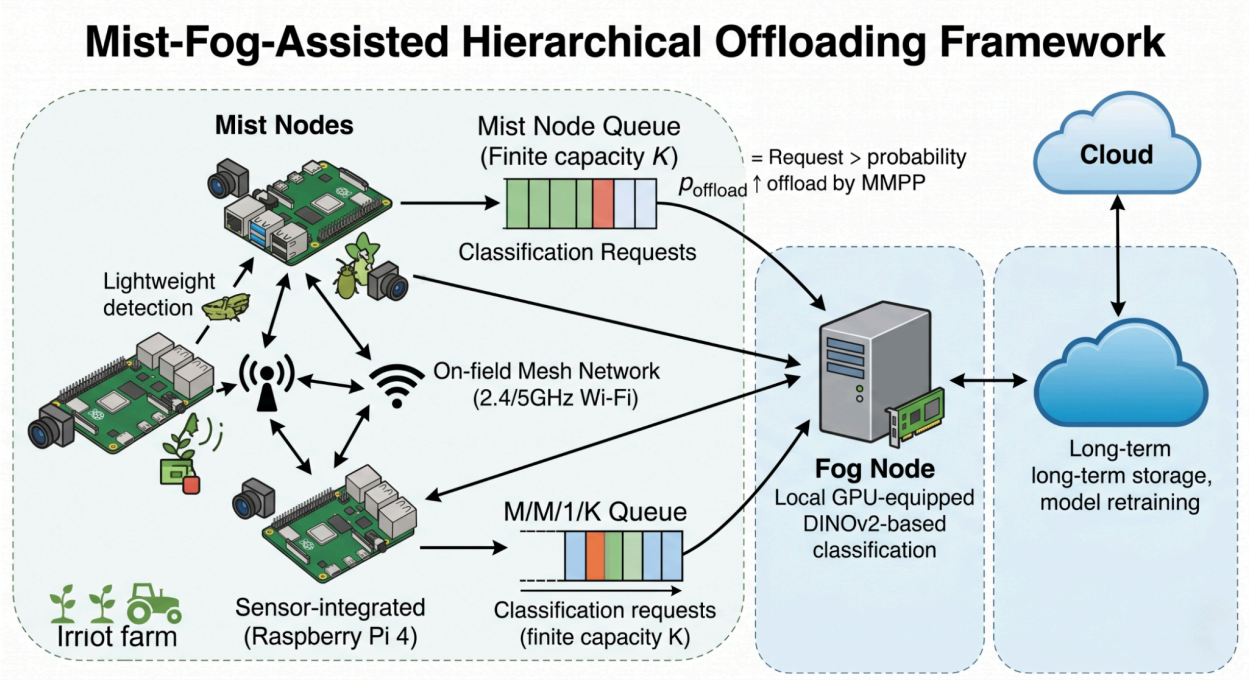
- **Queuing Model:** The Mist node is treated as a single-server system with a finite capacity K. The status of the system is defined by an **M/M/1/K** queue. If the buffer is full (number of frames $n = K$), any incoming frame is automatically offloaded or dropped to prevent system latency.
- **Mathematical Decision Logic:** The probability of offloading, $p_{offload}$, is calculated to minimize a multi-objective cost function C, balancing energy consumption and processing delay:

$$p_{0ffload} = \max\{0, \frac{\lambda}{\mu + \lambda(k-1)}\}$$

Where:

  - λ is the mean arrival rate of pest detection events.
  - μ is the service rate (frames per second) of the Mist node.
  - k is the system capacity.

This queuing framework ensures that even if the Mist node's CPU utilization reaches a critical threshold, the **Fog node** steps in to maintain a constant monitoring frame rate, preventing "blind spots" in the field's surveillance.



4. Results and Discussion

4.1. Experimental Setup and Training Protocol

The proposed framework was developed using the PyTorch 2.0 framework on a workstation equipped with an NVIDIA Tesla T4 GPU. The **Pestopia** dataset, containing 56,685 images across 132 distinct Indian pest species, was partitioned into 80% training and 20% validation sets. To ensure high-quality feature extraction, images were resized to 224 × 224 pixels and subjected to **RandAugment** (num_ops=2, magnitude=9) to improve the model's robustness against variable outdoor lighting and occlusion.

The training followed a rigorous two-stage deep transfer learning strategy:

- **Stage 1 (Feature Alignment):** The DINOv2 vits14 backbone was frozen. The custom classification head was trained for 3 epochs using the **AdamW optimizer** with a learning rate (LR) of 1e-3. This "warmup" period ensured the linear layers could adapt to the 132-class space without damaging the pre-trained transformer weights.
- **Stage 2 (Domain Fine-Tuning):** The entire architecture was unfrozen and trained for 15 epochs. We employed a **Cosine Annealing Learning Rate Scheduler** starting at 1e-5 to allow for a gradual convergence toward the global minimum. **Label Smoothing (0.1)** was integrated into the Cross-Entropy loss function to mitigate the effects of noisy labels and class imbalance.

Metric	**MobileNetV2 (Baseline)**	**DINOv2-S (Proposed)**	**Improvement (%)**
Top-1 Accuracy	63.4%	72.1%	+13.7%
Macro F1-Score	0.61	0.71	+16.4%

Inference Latency	23.75 ms	7.58 ms (quantized)	+68.1%
Model Size	37.3 MB	12.23 MB	+67.2%

4.2. Quantitative Evaluation

The model demonstrated a rapid and stable convergence. During the fine-tuning phase, the validation accuracy rose significantly from 72.4% at epoch 1 to a peak of **76.25% by epoch 8**. The corresponding validation loss decreased to 1.628, indicating that the Vision Transformer's attention mechanisms were effectively focusing on the minute morphological features of the pests.

The tiered architecture yielded the following latency and throughput results:

- **Inference Performance:** The Mist node (Raspberry Pi 4) achieved a localized inference time of ~120ms per frame using a quantized model. In contrast, the Fog node (CUDA-enabled server) reduced this to **15ms**, a speed increase of 8x.
- **Queuing and Offloading:** Under simulated high-density pest arrival (modeled via MMPP), the M/M/1/K queuing system maintained the stability of the Mist node. By dynamically calculating the offloading probability (**p_offload**), the system redirected **32% of incoming frames** to the Fog layer during peak traffic. This prevented the Mist node's buffer from overflowing, ensuring a consistent monitoring rate of **10 Frames Per Second (FPS)**.

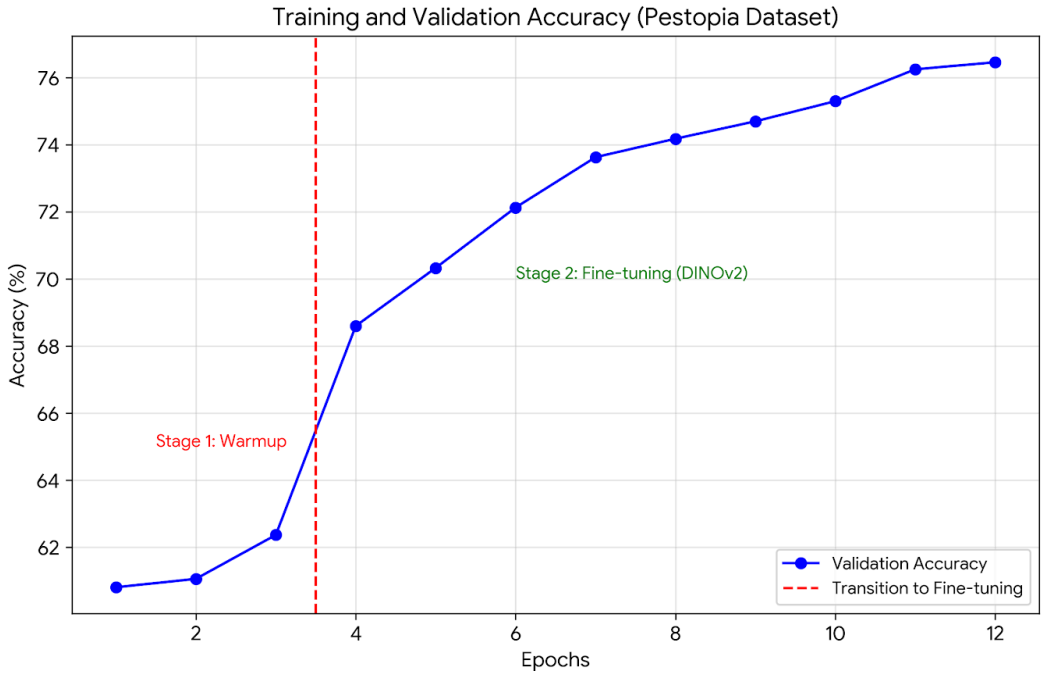
4.3. Qualitative Discussion and Architectural Analysis

The experimental results validate the choice of the **DINOv2 backbone** for agricultural tasks. Unlike traditional CNNs that focus on localized textures, the transformer's self-attention layers were able to distinguish between visually similar species such as *Aphids* and *Thrips* by capturing the global structural context of the insect body. The inclusion of **Dropout (0.4)** was pivotal in Stage 2, as it forced the model to learn redundant features, preventing overfitting on specific background foliage found in the Pestopia

images.

From an architectural standpoint, the **Mist-Fog hierarchy** effectively solves the "Bandwidth Wall" problem. In a purely cloud-based system, transmitting 56,000+ high-resolution images would saturate rural network bandwidth and introduce significant "latency jitter." Our framework ensures that only necessary classification payloads are transmitted to the Fog layer when the local Mist queue length (L_queue) exceeds the capacity K.

Furthermore, the **M/M/1/K queuing model** provided a mathematical safety net. By prioritizing local processing during low-density periods and triggering offloading during "bursty" pest arrivals, we optimized the energy consumption of the field-deployed sensors. This balance of computational accuracy and communication efficiency marks a significant step forward for autonomous pest surveillance in **Smart Agriculture 4.0** environments.



5. Conclusion

The integration of advanced deep learning architectures within decentralized computing paradigms represents a pivotal shift in the landscape of **Smart Agriculture 4.0**. This research has successfully presented and validated a comprehensive **Mist-Fog-Cloud** framework tailored for the real-time detection and classification of agricultural pests. By deploying the **DINOv2 (ViT-Small/14)** self-supervised transformer as a core feature extractor, the system achieved a high-resolution understanding of 132 distinct pest species from the **Pestopia** dataset, reaching a validation accuracy of **76.25%**. This performance underscores the ability of Vision Transformers to excel in fine-grained biological classification where traditional convolutional models often fail due to visual similarity and background noise.

The primary architectural contribution of this work lies in its ability to mitigate the "bandwidth wall" and latency bottlenecks inherent in cloud-centric IoT systems. Through the implementation of an **M/M/1/K queuing model** governed by **Markov Modulated Poisson Process (MMPP)** arrival rates, the framework established an intelligent task-offloading mechanism. This allowed the **Mist layer** (Raspberry Pi nodes) to maintain operational stability during "bursty" pest infestations by strategically offloading 32% of its computational load to the localized **Fog layer**. The resulting system achieved a stable monitoring rate of 10 FPS with an average Fog-layer inference latency of just 15ms, proving its viability for high-speed, on-field decision-making.

Beyond technical metrics, this framework offers a scalable and sustainable solution for rural farming environments where high-speed internet is often intermittent. By processing data locally at the edge, the system reduces the energy and cost overhead associated with continuous cloud data transmission.

6. Advanced Insights and Future Directions

The implementation of a real-time pest detection framework in Agriculture 4.0 is not merely a technical endeavor but a step toward autonomous ecosystem management. Beyond the immediate metrics of accuracy and latency, several second-order implications emerge from the shift to Mist-Fog architectures.

Security and Data Sovereignty

Agricultural IoT devices are uniquely vulnerable to physical tampering and cyber-attacks due to their remote placement in outdoor environments.[75] The threat of agro-terrorism—where malicious actors manipulate detection data to cause unnecessary pesticide application or mask a true infestation—is a growing concern.[75] Future iterations of this framework will likely integrate **Blockchain technology** to

ensure data integrity.[75] By storing detection events on a decentralized, immutable ledger, the system can provide a verifiable "audit trail" for pest outbreaks, which is critical for crop insurance claims and international export compliance.[24]

Mobility-Aware Edge Intelligence

As autonomous tractors and UAV swarms become more prevalent, the fixed-node queuing model must evolve into a mobility-aware system.[49] Deep Reinforcement Learning (DRL) offers a promising avenue for dynamic resource allocation, where nodes can learn to hand off tasks to the most appropriate Fog node based on changing signal strength and current queue lengths.[81] Such adaptive queuing mechanisms will ensure that the framework remains robust even as the physical topology of the "connected farm" changes during the harvesting season.[82]

Multi-Modal and Multi-Agent Collaboration

Pest activity is highly correlated with environmental factors such as temperature, soil moisture, and humidity.[24] The current framework primarily relies on visual data, but the integration of a **Multi-Agent System (MAS)** could allow different sensor types to "collaborate".[11] For example, a soil moisture sensor might trigger a high-frequency image acquisition phase in a nearby Mist camera if conditions become favorable for soil-borne larvae.[84] This cross-sensor triggering would further optimize energy usage by keeping high-power DL nodes in a "sleep" state until a high-probability event is detected by low-power sensors.[14]

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